

$x_0=0, f(x_0)$
 $x_1=1, f(x_1)$
 $x_2=2, f(x_2), f'(x_2)$ (Hermite)

$$J_0 = \{0\}$$

$$J_1 = \{1\}$$

$$J_2 = \{0, 2\}$$

$n=2$ index of point

$$m = 1 + 1 + 2 - 1 = 3$$

↳ degree of Birkhoff

$$B_m f(x) = \sum_{k=0}^m \sum_{j \in J_k} b_{kj}(x) f^{(j)}(x_k)$$

$$B_3 f(x) = b_{00}(x) f(0) + b_{11}(x) f(x_1) + b_{20}(x) f(x_2) + b_{22}(x) f'(x_2)$$

$$b_{00}(x) = ax^3 + bx^2 + cx + d$$

$$b_{00}'(x) = 3ax^2 + 2bx + c$$

$$b_{00}''(x) = 6ax + 2b$$

$$\begin{cases} b_{00}(x_0) = 1 \\ b_{00}'(x_1) = 0 \\ b_{00}(x_2) = 0 \\ b_{00}'(x_2) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} d = 1 \\ 3a \cdot 1 - 12a \cdot 1 + c = 0 \\ a \cdot 2^3 + b \cdot 2^2 + 2c + 0 = 0 \\ 6a \cdot 2 + 2b = 0 \\ \Rightarrow 12a + 2b = 0 \\ 6a + b = 0 \\ b = -6a \end{cases}$$

$$\Rightarrow \begin{cases} d = 1 \\ c = 9a \\ 8a - 24a + 12a + 1 = 0 \\ b = -6a \end{cases}$$

$$\Rightarrow \begin{cases} 26a - 24a = 1 \Rightarrow 2a = 1 \end{cases}$$

$$\Rightarrow \begin{cases} a = \frac{1}{2} \\ b = -3 \\ c = \frac{9}{2} \\ d = 1 \end{cases}$$

b₀₀

② $f: [0,1] \rightarrow \mathbb{R}$ with continuous 3rd order derivatives
 a) Find quadrature formula w. max. degree of precision,
 of the form: $\int_0^1 f(x) dx = A \cdot f(0) + B \cdot f'(x_1) + R(f)$
 b) $f(0), f'(1/3) \Rightarrow P_f(x)$? Birkhoff
 c) $\int_0^1 (P_f(x)) dx$
 d) $R(f)$

$$a) R(f) = \int_0^1 f(x) dx - A \cdot f(0) - B \cdot f'(x_1)$$

$$f(x) = \frac{1}{2}x^3 - 3x^2 + \frac{9}{2}x + 1$$

$$a) R(f) = \int_0^1 f(x) dx - A \cdot f(0) - B \cdot f'(x_1)$$

$$R(e_0) = \int_0^1 e_0(x) dx - A \cdot e_0(0) - B \cdot e_0'(x_1)$$

$$= x \Big|_0^1 - A \cdot 1 - B \cdot 0$$

$$1 - A = 0 \Rightarrow A = 1$$

$$R(e_1) = \int_0^1 e_1(x) dx - 1 \cdot e_1(0) - B \cdot e_1'(x_1)$$

$$R(e_1) = \frac{x^2}{2} \Big|_0^1 - 0 - B = \frac{1}{2} - B$$

$$\frac{1}{2} - B = 0 \Rightarrow B = \frac{1}{2}$$

$$R(e_2) = \int_0^1 e_2(x) dx - 1 \cdot e_2(0) - \frac{1}{2} e_2'(x_1)$$

$$= \frac{x^3}{3} \Big|_0^1 - 0 - x_1 = \frac{1}{3} - x_1$$

$$\Rightarrow x_1 = \frac{1}{3}$$

$$R(e_3) = \int_0^1 e_3(x) dx - 1 \cdot e_3(0) - \frac{1}{2} e_3'(x_1)$$

$$= \frac{x^4}{4} \Big|_0^1 - 0 - \frac{1}{2} \cdot \frac{3}{9} = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$$

$$(x,)$$

$$)$$

$$(x,)$$

$$x_1 = 0$$

$$\frac{1}{3})$$

$$\neq 0$$

degree of maximum precision is 2
 for $A = 1, B = \frac{1}{2}, x_1 = \frac{1}{3}$

b) $f(0), f'(\frac{1}{3})$ $\Rightarrow$ $P_f(x)$?

$$I_0 = \{0\} \quad \begin{matrix} x_0 = 0 \\ x_1 = \frac{1}{3} \end{matrix}$$

$$I_1 = \{1\}$$

$$\Rightarrow n = 1 + 1 - 1_m = 1$$

$$B_1 f(x) = \sum_{k=0}^n \sum_{j \in J_k} b_{kj}(x) f^{(j)}(x_k)$$

$$B_1 f(x) = b_{00}(x) f(x_0) + b_{11}(x) f'(\frac{1}{3})$$

$$b_{00}(x) = ax + b$$

$$b_{00}'(x) = a$$

$$\begin{cases} b_{00}(x_0) = 1 \\ b_{00}'(x_1) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} a \cdot 0 + b = 1 \\ a = 0 \end{cases}$$

$$\begin{cases} b_{11}(x_0) = 0 \\ b_{11}'(x_1) = 1 \end{cases}$$

$$\Rightarrow \begin{cases} a \cdot 0 + b = 0 \\ a = 1 \end{cases}$$

$$f^{(1)} x_k$$

(1)

$$\left. \begin{array}{l} a = 1 \\ b = 0 \end{array} \right\} \Rightarrow b_{00}(x) = 1$$

$$\left. \begin{array}{l} a = 0 \\ b = 1 \end{array} \right\} \Rightarrow b_{11}(x) = x$$

$$f'(x,)$$

$$B_1 f(x) = 1 \cdot f(x_0) + x \cdot f'(x_0)$$

$$= f(0) + x \cdot f'(0)$$

$$c) \int_0^1 f(0) + x \cdot f'(\frac{1}{3}) dx = f(0)$$

$$= f(0) + \frac{1}{2} f'(\frac{1}{3})$$

$$d) f(x) = \cancel{B_1 f(x)} + R_1 f(x)$$

$$R_1 f(x) = f(x) - B_1 f(x)$$

$$Rf = \int_0^1 f(x) dx - \int_0^1 B_1 f(x) dx$$

$$Rf = \int_0^1 f(x) dx - f(0) + \frac{1}{2} f'(0)$$

$$R_1 f(x) = f(x) - B_1 f(x)$$

$$f'(x_1)$$

$$f'(\frac{1}{3})$$

$$0) \cdot x|_0 + f'(\frac{1}{3}) \cdot \frac{x^2}{2}|_0$$

— creare pe polinoame

$$\left. \begin{array}{l} 0) dx \\ \frac{1}{2} f'(\frac{1}{3}) \end{array} \right\} \text{fără aster}$$

$$1)$$

$$1, 1, 1, 1$$

$$R_1(f(x)) = \int_0^1 K_2(x, t) f(t) dt$$

$$K_2(x, t) = \frac{1}{2!} R_1((x-t)_+^2)$$

$$R_1((x-t)_+^2) = (x-t)_+^2 - \underbrace{(0-t)_+^2}_{=0}$$

$$B_1(f(x)) = f(0) + x \cdot f'(\frac{1}{3})$$

$$\text{I } R_1((x-t)_+^2) = (x-t)_+^2 - 0$$

$$= \underbrace{(x-t)_+^2}_{\geq 0} + \underbrace{2x}_{\geq 0} \underbrace{(\frac{1}{3}-t)}_{\geq 0}$$

$$\Rightarrow Rf = \frac{1}{2!} f''(\xi)$$

$$y'(t) dt$$

$$\left. \frac{t^2}{2} + x \frac{\partial}{\partial x} (x-t)^2 \right|_{x=\frac{1}{3}}$$

$$+ x \cdot 2(x-t) \Big|_{x=\frac{1}{3}}$$

$$\frac{t}{2} \neq \Rightarrow \text{Peano Theorem}$$

$$Re_2$$

$$1 \cap \rightarrow 1 p^M(t) (R_p$$

$$\int_0^1 R f \approx \int_0^1 \frac{1}{2} f''(\xi) \approx \frac{1}{2} f''(\xi)$$

③ Gaussian elim. w partial pivot

$$\begin{bmatrix} -1 & 3 & 2 \\ 2 & 4 & 0 \\ 1 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \\ 7 \end{bmatrix}$$

$$A = \left[\begin{array}{ccc|c} -1 & 3 & 2 & -1 \\ 2 & 4 & 0 & 8 \\ 1 & 2 & -3 & 7 \end{array} \right]$$

$$\approx \left[\begin{array}{ccc|c} 2 & 4 & 0 & 8 \\ 0 & 5 & 2 & 3 \\ 0 & 0 & -2 & 2 \end{array} \right]$$

$$) Re_3 \approx \frac{1}{2} f'''(\xi) > 0 Re_3$$

$$\frac{1}{12} \approx \frac{1}{24} f'''(\xi)$$

$$R_1 \leftrightarrow R_1,$$

$$\approx A \begin{bmatrix} 2 & 4 & 0 & 12 \\ -1 & 3 & 2 & -1 \\ 1 & 2 & -3 & 7 \end{bmatrix} =$$

$$R_2 \leftarrow R_2 + \frac{1}{2} R_1$$

$$R_3 \leftarrow R_3 - \frac{1}{2} R_1$$

3

- 1

2/

$$\begin{pmatrix} 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} 3 \end{pmatrix}$$

$$\begin{cases} 2x_1 + 4x_2 = \\ 5x_2 + 2x_3 = \\ -3x_3 = \end{cases}$$

$$X' = \begin{bmatrix} 2 & 1 & -1 \end{bmatrix}$$

$$\textcircled{1} \quad g(x) = \frac{1 - \sin x}{2}$$

a) show that $x = g(x)$ has unique solution

b) Find $[a, b]$ such that $\alpha \in [a, b] \Rightarrow$
 converges to α for any $x_0 \in [a, b]$
 + order convergence + error

$$\begin{matrix} 8 \\ z=3 \\ z=3 \\ J^T \end{matrix}$$

$$z)$$

$$\left\{ \begin{array}{l} \\ \\ \\ \end{array} \right.$$

$$x_1 z$$

$$x_2 z$$

$$x_3 z$$

sol. α

$$x_{m+1} = \frac{1 - \sin x_m}{2}$$

, b]

2

1

1